GIT AND GITHUB TASK

1. Create a GitHub account.

Step 1: Open GitHub

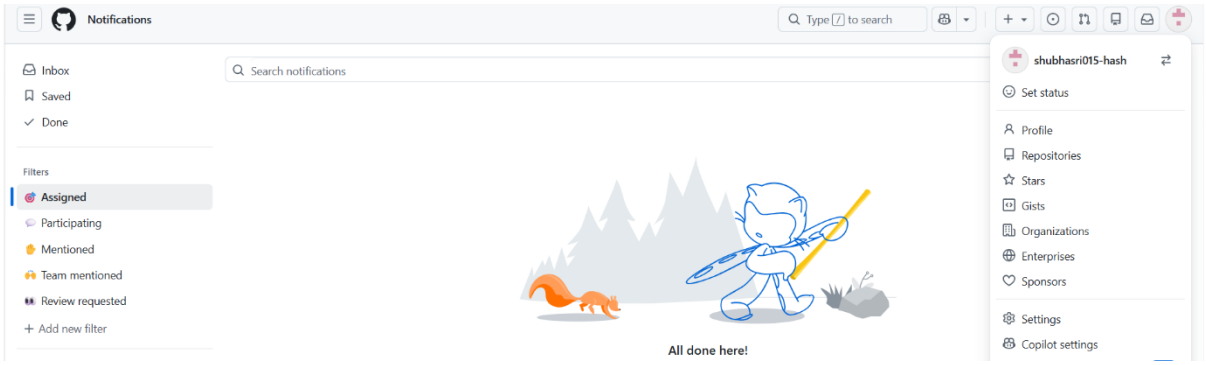
-Go to the official website: https://github.com/

-Click Sign up

-Enter your details

-Verify your email

-Log in



2. Set up MFA (Multi-Factor Authentication) for the GitHub account using a mobile app.

Steps :

-Step 1: Log in to GitHub

-Settings → Password and authentication.

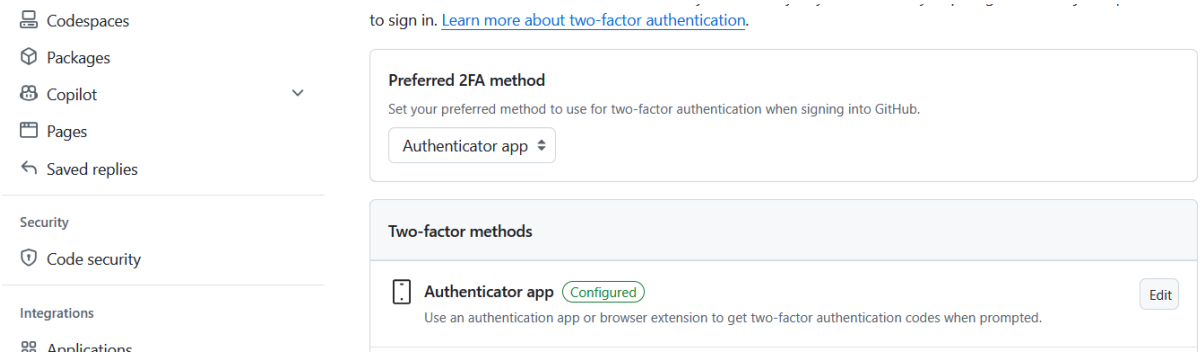
-Enable MFA

Under Two-factor authentication, click Enable two-factor authentication and choose the authenticator app option.

-Install a Microsoft authenticator mobile app

-Step 5: Enter the verification code

-Step 6: Save recovery codes



3.Execute all commands shown in the video

Git version -
Checks whether Git is installed on your computer and displays its version.
Git status -Shows the current state of your repository. The Dockerfile appears as an untracked file because you have not added it to Git yet.
git init -Creates a hidden .git directory to start tracking your project with Git.
If you see Initialized empty Git repository, the command worked successfully.
Ls -a =You should see .git and Dockerfile.

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop
$ git status
fatal: not a git repository (or any of the parent directories): .git

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop
$ git init
Initialized empty Git repository in C:/Users/Shubhasri/OneDrive/Desktop/.git/

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop (master)
$ ls -l *dockerfile*
-rw-r--r-- 1 Shubhasri 197121 0 Oct 22 15:34 dockerfile

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop (master)
$ mkdir hiring-app

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop (master)
$ cd hiring-app

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ git init
Initialized empty Git repository in C:/Users/Shubhasri/OneDrive/Desktop/hiring-app/.git/

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ touch Dockerfile

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ ls
Dockerfile

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ git init
Reinitialized existing Git repository in C:/Users/Shubhasri/OneDrive/Desktop/hiring-app/.git/

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ ls -a
./  ../  .git/  Dockerfile

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ cd .git/

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app/.git (GIT_DIR!)
$ cd ..

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ git status
On branch master

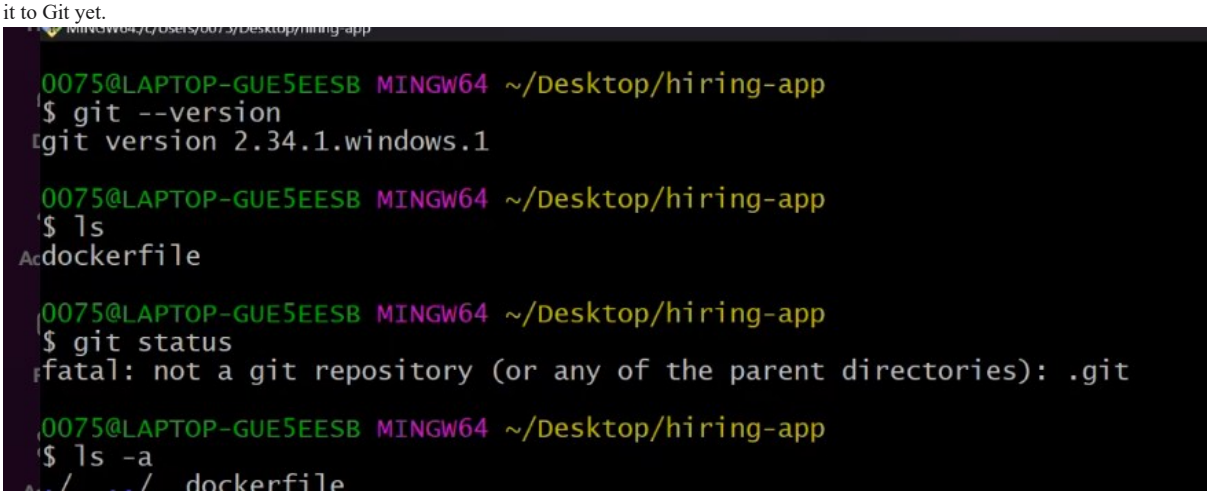
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile

nothing added to commit but untracked files present (use "git add" to track)

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ git config --list
diff.astextplain.textconv=astextplain
```

4 .Git status -Shows the current state of your repository. The Dockerfile appears as an untracked file because you have not added it to Git yet.



5.Git config –list : is used to display all Git configuration settings.

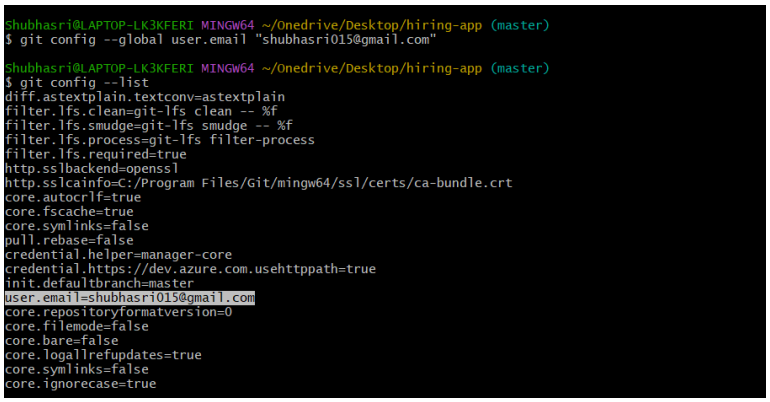
```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager-core
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$ git --version
git version 2.37.1.windows.1

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/Onedrive/Desktop/hiring-app (master)
$
```

6.Git config -- global user.name “your email address” Sets the name Git records as the author of your commits.

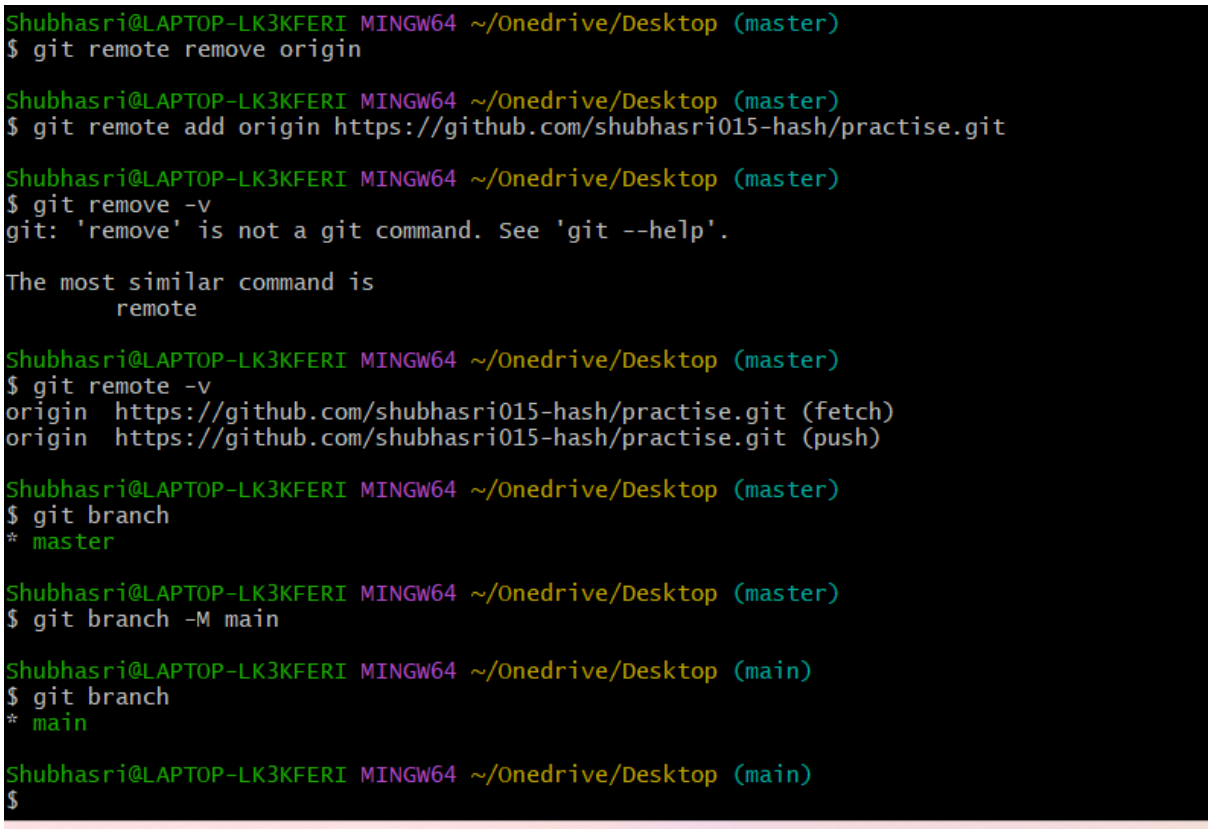
Git config --list = Displays all applicable Git settings



7.git remote remove origin is a Git command used to remove the remote connection named origin from your local Git repository.

git branch

git branch --show-current git branch -M main

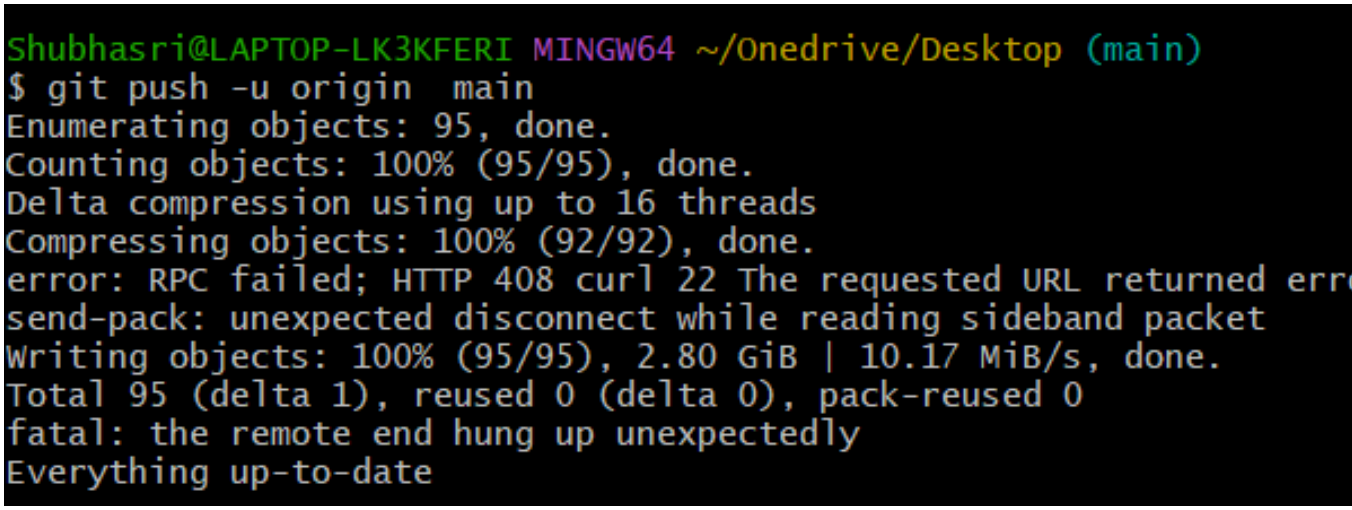


8. Push code to GitHub

Ssh -keygen

Git push -u origin main





9.Pull changes from GitHub

git pull

I used git pull to download the latest changes from the remote GitHub repository and integrate those changes into my current local branch.

Fetch command

I used git fetch to download information about changes from the remote repository without automatically merging those changes into my current branch

Clone :

I used git clone when I wanted to get an existing GitHub repository onto my local machine. It downloads the repository, including its files and Git history**.**

```
Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/subh/trayaksh (main)
$ git push
Everything up-to-date

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/subh/trayaksh (main)
$ git pull
Already up to date.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/subh/trayaksh (main)
$ git pull
Already up to date.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/subh/trayaksh (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/subh/trayaksh (main)
$ git fetch

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/subh/trayaksh (main)
$ git fetch origin

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/subh/trayaksh (main)
$ git clone git@github.com:shubhasri015-hash/task1.git
Cloning into 'task1'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

Shubhasri@LAPTOP-LK3KFERI MINGW64 ~/subh/trayaksh (main)
$ cd task1
```

Revert command:

I first used git log --oneline to find the commit ID. Then I used git revert

<commit-id> to undo the changes introduced by that commit. Git creates a new commit to reverse the previous changes**.**

